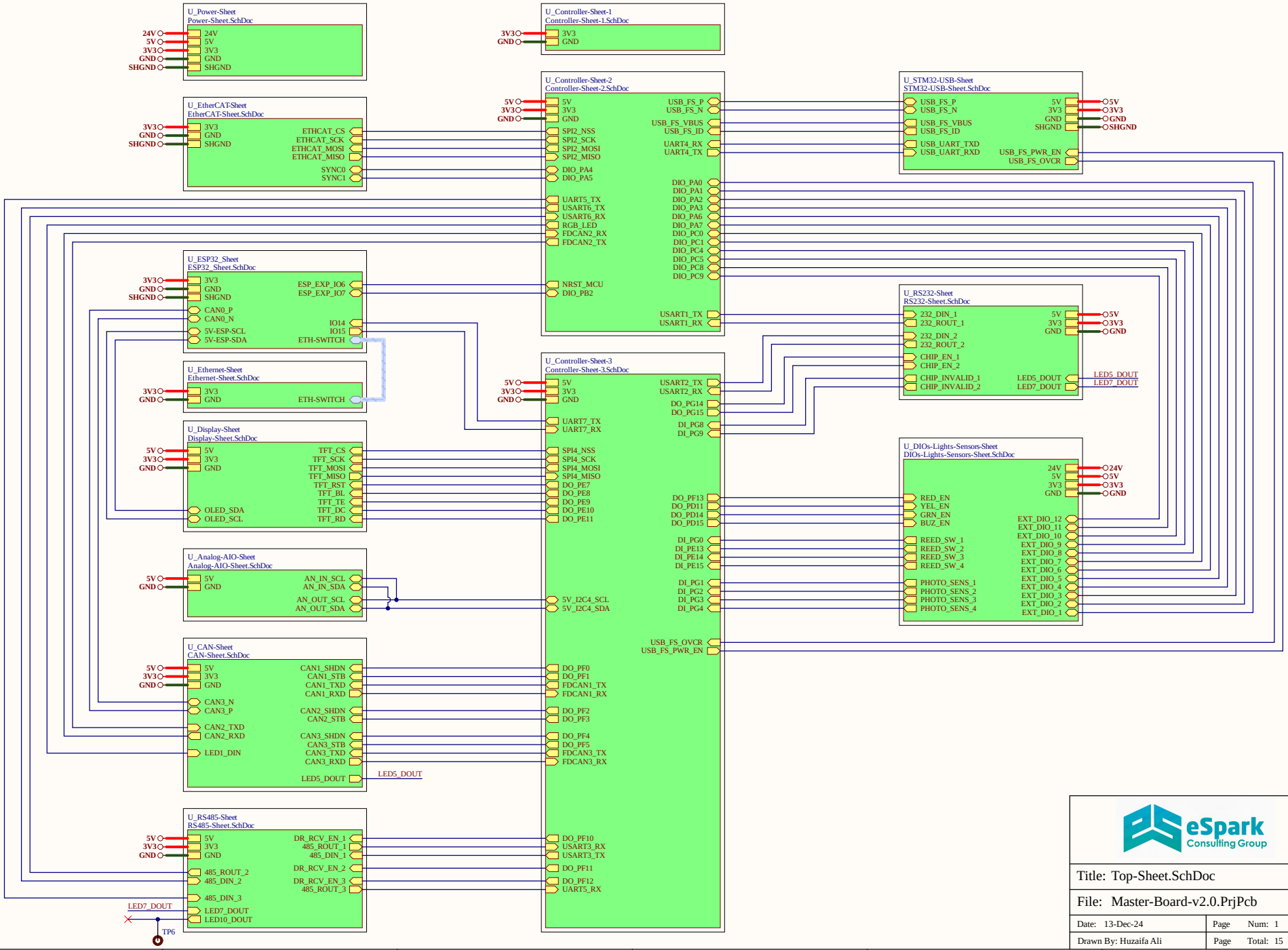
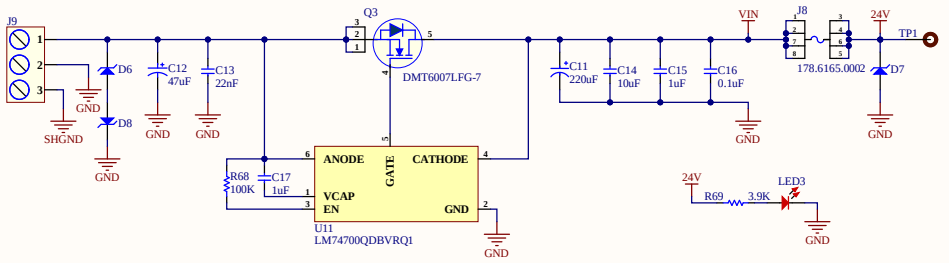
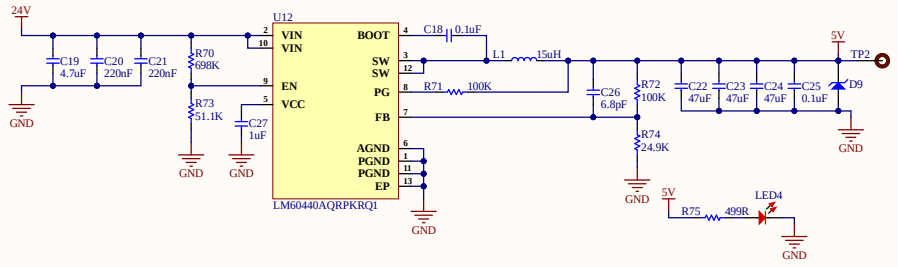

U_Power-Sheet
Power-Sheet.SchDoc
24V
5V
3V3
GND
SHGND
U_Controller-Sheet-1
Controller-Sheet-1.SchDoc
3V3
GND
U_EtherCAT-Sheet
EtherCAT-Sheet.SchDoc
ETHCAT_CS
ETHCAT_SCK
ETHCAT_MOSI
ETHCAT_MISO
SYNC0
SYNC1
U_Controller-Sheet-2
Controller-Sheet-2.SchDoc
USB_FS_P
USB_FS_N
USB_FS_VBUS
USB_FS_ID
UART4_RX
UART4_TX
SPI2_NSS
SPI2_SCK
SPI2_MOSI
SPI2_MISO
DIO_PA4
DIO_PA5
DIO_PA0
DIO_PA1
DIO_PA2
DIO_PA3
DIO_PA6
DIO_PA7
DIO_PC0
DIO_PC1
DIO_PC4
DIO_PC5
DIO_PC8
DIO_PC9
UART5_TX
USART6_TX
USART6_RX
RGB_LED
FDCAN2_RX
FDCAN2_TX
NRST_MCU
DIO_PB2
USART1_TX
USART1_RX
U_STM32-USB-Sheet
STM32-USB-Sheet.SchDoc
USB_FS_P
USB_FS_N
USB_FS_VBUS
USB_FS_ID
USB_UART_TXD
USB_UART_RXD
USB_FS_PWR_EN
USB_FS_OVCR
SHGND
U_ESP32_Sheet
ESP32_Sheet.SchDoc
ESP_EXP_IO6
ESP_EXP_IO7
CAN0_P
CAN0_N
5V-ESP-SCL
5V-ESP-SDA
IO14
IO15
ETH-SWITCH
U_Ethernet-Sheet
Ethernet-Sheet.SchDoc
ETH-SWITCH
U_Display-Sheet
Display-Sheet.SchDoc
TFT_CS
TFT_SCK
TFT_MOSI
TFT_MISO
TFT_RST
TFT_BL
TFT_TE
TFT_DC
TFT_RD
OLED_SDA
OLED_SCL
U_Analog-AIO-Sheet
Analog-AIO-Sheet.SchDoc
AN_IN_SCL
AN_IN_SDA
AN_OUT_SCL
AN_OUT_SDA
5V_I2C4_SCL
5V_I2C4_SDA
U_CAN-Sheet
CAN-Sheet.SchDoc
CAN1_SHDN
CAN1_STB
CAN1_TXD
CAN1_RXD
CAN2_SHDN
CAN2_STB
CAN3_SHDN
CAN3_STB
CAN3_TXD
CAN3_RXD
CAN3_N
CAN3_P
CAN2_TXD
CAN2_RXD
LED1_DIN
LED5_DOUT
U_RS485-Sheet
RS485-Sheet.SchDoc
DR_RCV_EN_1
485_ROUT_1
485_DIN_1
DR_RCV_EN_2
DR_RCV_EN_3
485_ROUT_3
485_ROUT_2
485_DIN_2
485_DIN_3
LED7_DOUT
LED10_DOUT
TP6
U_Controller-Sheet-3
Controller-Sheet-3.SchDoc
USART2_TX
USART2_RX
DO_PG14
DO_PG15
DI_PG8
DI_PG9
UART7_TX
UART7_RX
SPI4_NSS
SPI4_SCK
SPI4_MOSI
SPI4_MISO
DO_PE7
DO_PE8
DO_PE9
DO_PE10
DO_PE11
DO_PF13
DO_PD11
DO_PD14
DO_PD15
DI_PG0
DI_PE13
DI_PE14
DI_PE15
DI_PG1
DI_PG2
DI_PG3
DI_PG4
USB_FS_OVCR
USB_FS_PWR_EN
DO_PF0
DO_PF1
FDCAN1_TX
FDCAN1_RX
DO_PF2
DO_PF3
DO_PF4
DO_PF5
FDCAN3_TX
FDCAN3_RX
DO_PF10
USART3_RX
USART3_TX
DO_PF11
DO_PF12
UART5_RX
U_RS232-Sheet
RS232-Sheet.SchDoc
232_DIN_1
232_ROUT_1
232_DIN_2
232_ROUT_2
CHIP_EN_1
CHIP_EN_2
CHIP_INVALID_1
CHIP_INVALID_2
LED5_DOUT
LED7_DOUT
U_DIOs-Lights-Sensors-Sheet
DIOs-Lights-Sensors-Sheet.SchDoc
RED_EN
YEL_EN
GRN_EN
BUZ_EN
REED_SW_1
REED_SW_2
REED_SW_3
REED_SW_4
PHOTO_SENS_1
PHOTO_SENS_2
PHOTO_SENS_3
PHOTO_SENS_4
EXT_DIO_12
EXT_DIO_11
EXT_DIO_10
EXT_DIO_9
EXT_DIO_8
EXT_DIO_7
EXT_DIO_6
EXT_DIO_5
EXT_DIO_4
EXT_DIO_3
EXT_DIO_2
EXT_DIO_1
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Consulting Group
Title: Top-Sheet.SchDoc
File: Master-Board-v2.0.PrjPcb
Date: 13-Dec-24
Page Num: 1
Drawn By: Huzaifa Ali
Page Total: 15

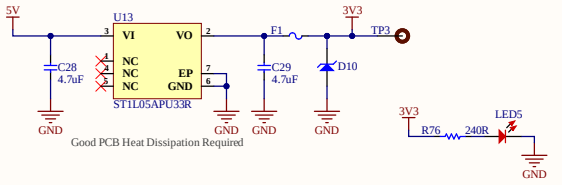
24V INPUT

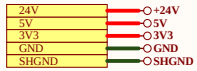


5V BUCK CONVERTER



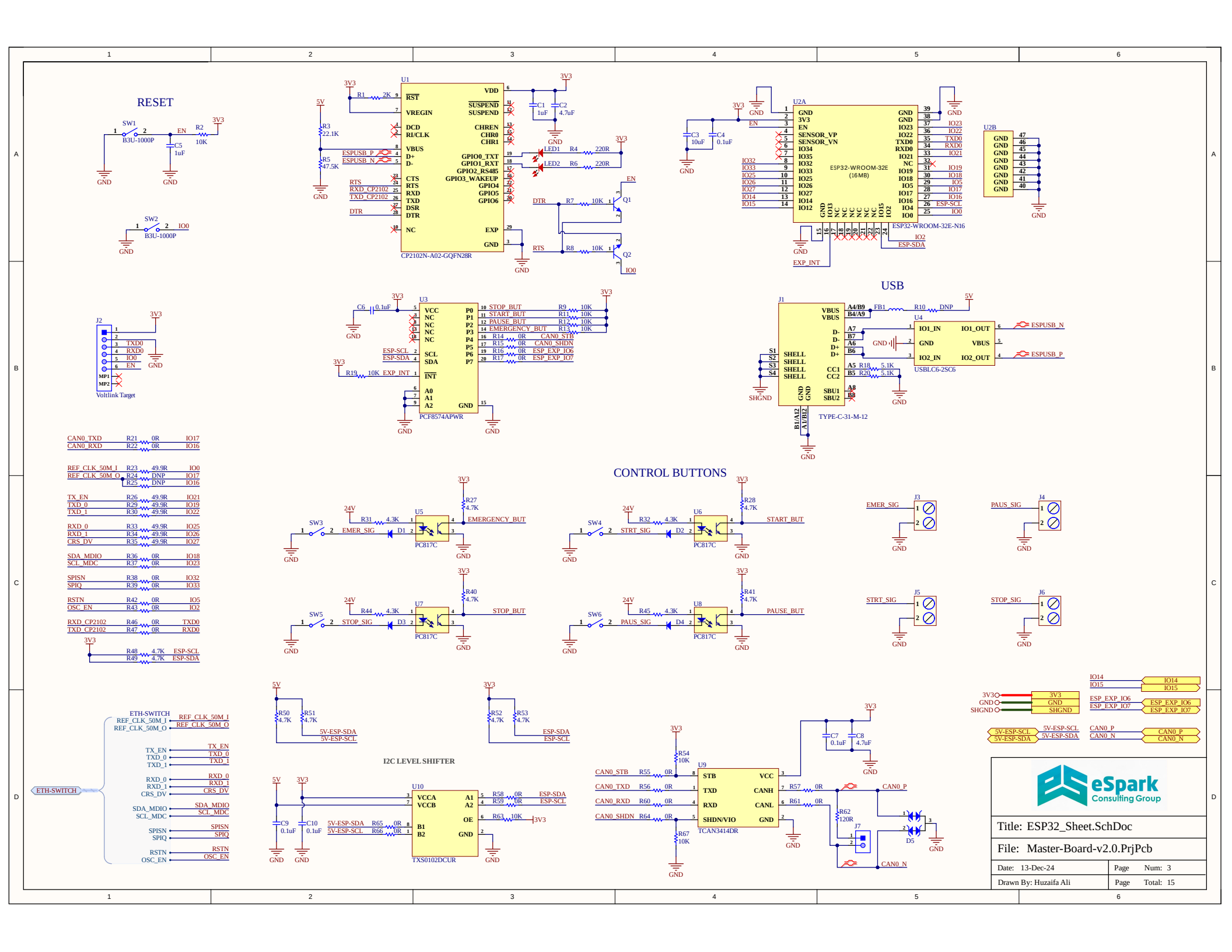
3.3V LINEAR REGULATOR





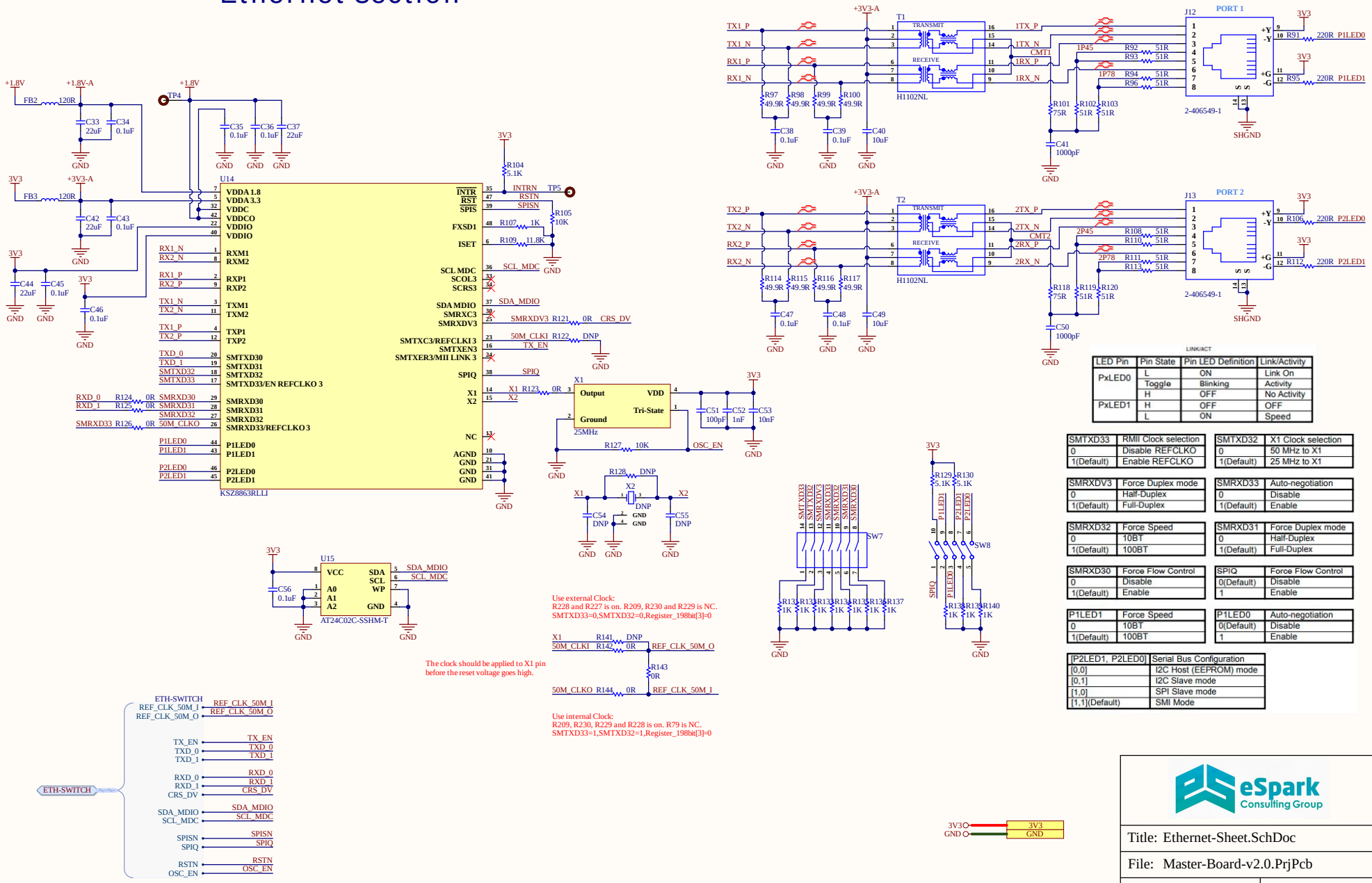
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Title: Power-Sheet.SchDoc	
File: Master-Board-v2.0.PrjPcb	
Date: 13-Dec-24	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 15



Title:	ESP32_Sheet.SchDoc
File:	Master-Board-v2.0.PrjPcb
Date: 13-Dec-24	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 15

Ethernet Section



Use external Clock:
R228 and R227 is on. R209, R230 and R229 is NC.
SMTXD33=0,SMTXD32=0,Register_198bit[3]=0

The clock should be applied to X1 pin before the reset voltage goes high.

Use internal Clock:
R209, R230, R229 and R228 is on. R79 is NC.
SMTXD33=1,SMTXD32=1,Register_198bit[3]=0

LINK/ACT

LED Pin	Pin State	Pin LED Definition	Link/Activity
PxLED0	L	ON	Link On
	Toggle	Blinking	Activity
	H	OFF	No Activity
PxLED1	H	OFF	OFF
	L	ON	Speed

SMTXD33	RMII Clock selection
0	Disable REFCLKO
1(Default)	Enable REFCLKO

SMTXD32	X1 Clock selection
0	50 MHz to X1
1(Default)	25 MHz to X1

SMRXDV3	Force Duplex mode
0	Half-Duplex
1(Default)	Full-Duplex

SMRXD33	Auto-negotiation
0	Disable
1(Default)	Enable

SMRXD32	Force Speed
0	10BT
1(Default)	100BT

SMRXD31	Force Duplex mode
0	Half-Duplex
1(Default)	Full-Duplex

SMRXD30	Force Flow Control
0	Disable
1(Default)	Enable

SPIQ	Force Flow Control
0(Default)	Disable
1	Enable

P1LED1	Force Speed
0	10BT
1(Default)	100BT

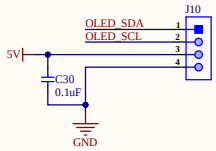
P1LED0	Auto-negotiation
0(Default)	Disable
1	Enable

[P2LED1, P2LED0]	Serial Bus Configuration
[0,0]	I2C Host (EEPROM) mode
[0,1]	I2C Slave mode
[1,0]	SPI Slave mode
[1,1](Default)	SMI Mode

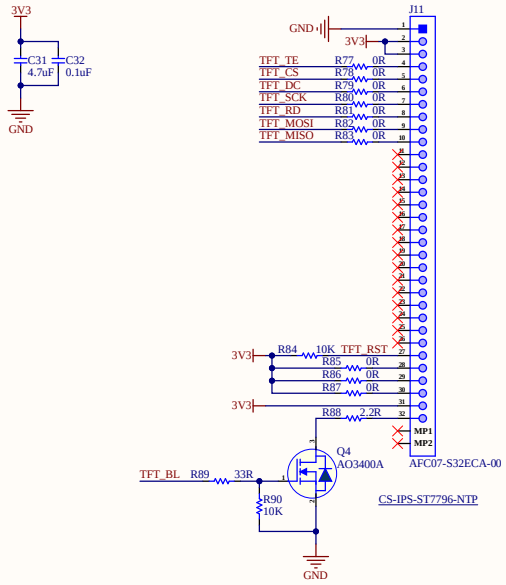
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File:	Master-Board-v2.0.PrjPcb
Date: 13-Dec-24	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 15

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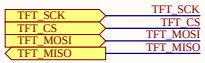
OLED DISPLAY

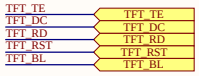


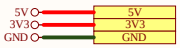
TFT DISPLAY







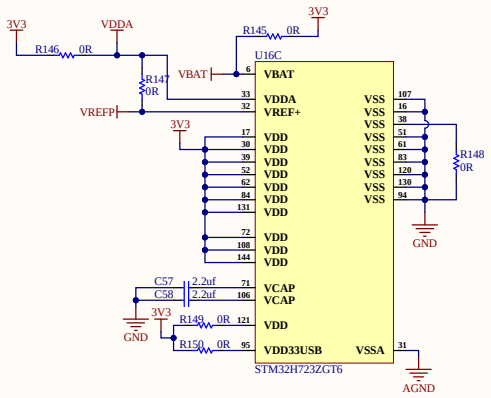


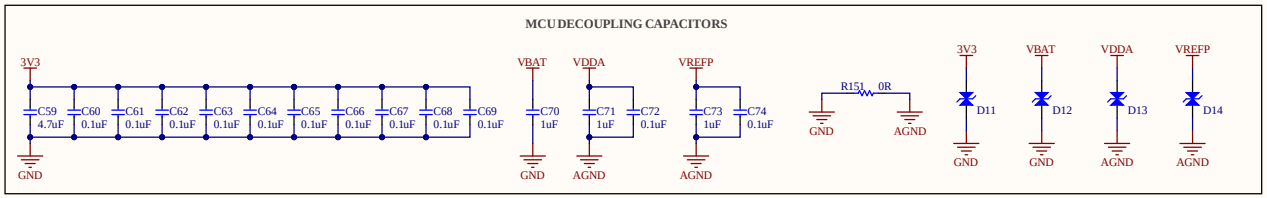


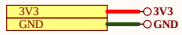
eSpark Consulting Group

Title: Display-Sheet.SchDoc	
File: Master-Board-v2.0.PrjPcb	
Date: 13-Dec-24	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 15

CONTROLLER POWER SHEET



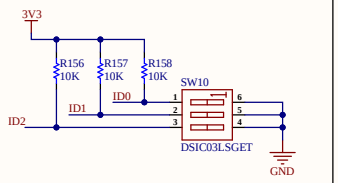




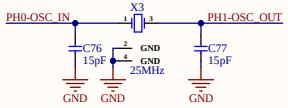
eSpark Consulting Group

Title: Controller-Sheet-1.SchDoc	
File: Master-Board-v2.0.PrjPcb	
Date: 13-Dec-24	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 15

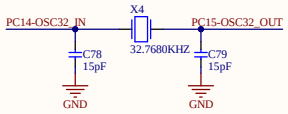
CONTROLLER IO SHEET 1

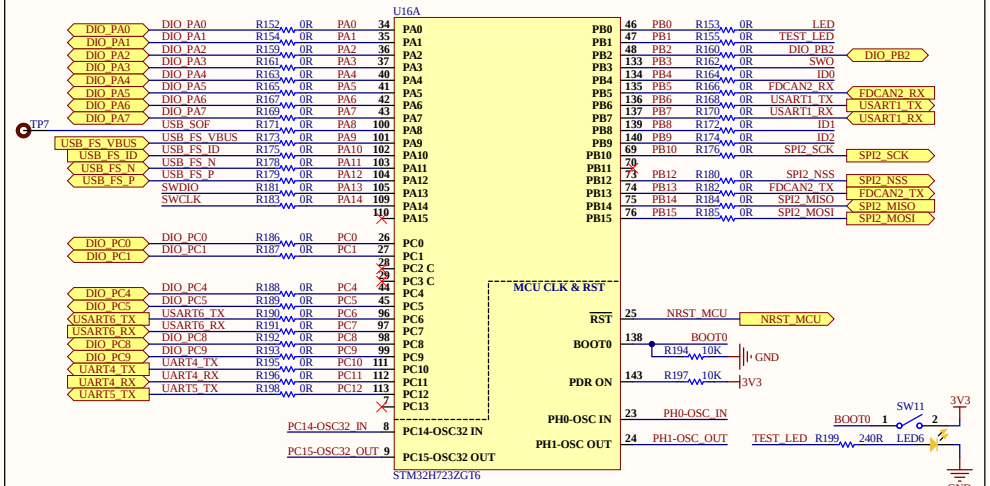


EXTERNAL HSE CLOCK

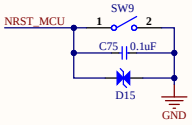


EXTERNAL LSE CLOCK

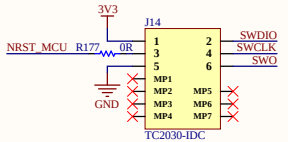


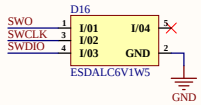


RESET FUNCTION

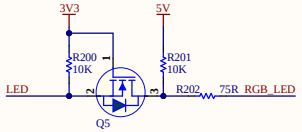


PROGRAMMING TAG CONNECT





RGB LED LEVEL TRANSLATOR



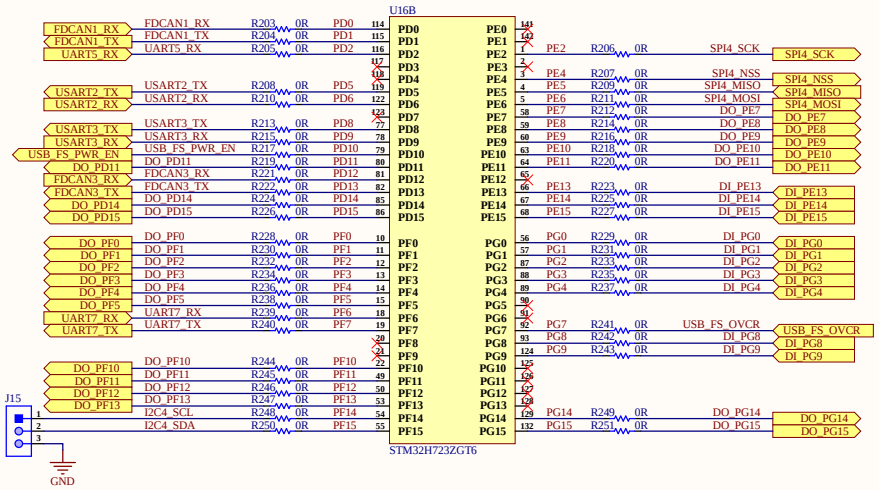




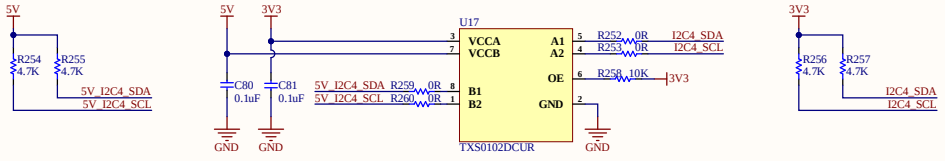
eSpark Consulting Group

Title:	Controller-Sheet-2.SchDoc
File:	Master-Board-v2.0.PrjPcb
Date: 13-Dec-24	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 15

CONTROLLER IO SHEET 2



I2C LEVEL SHIFTER



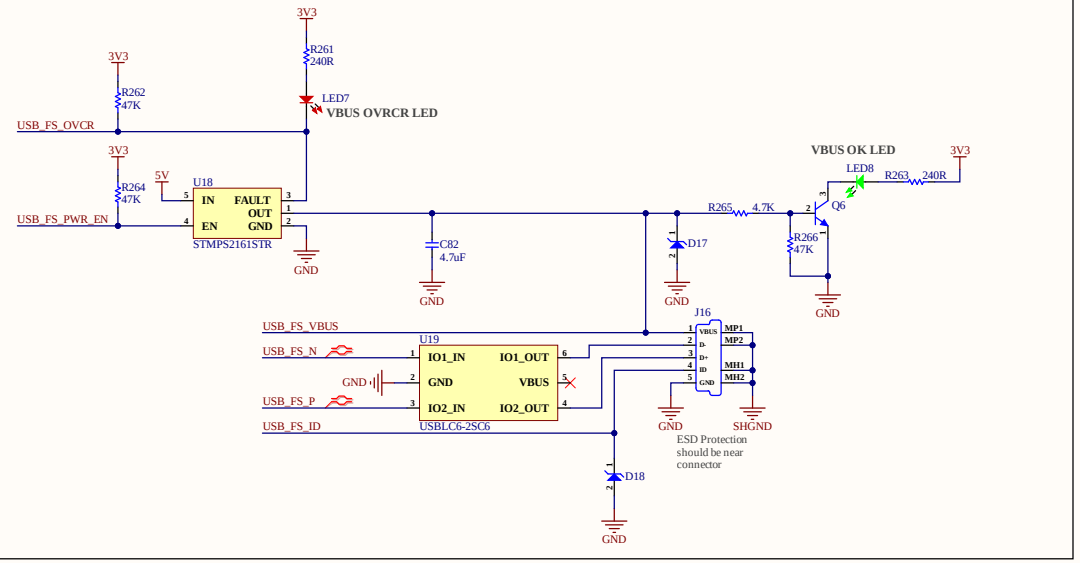




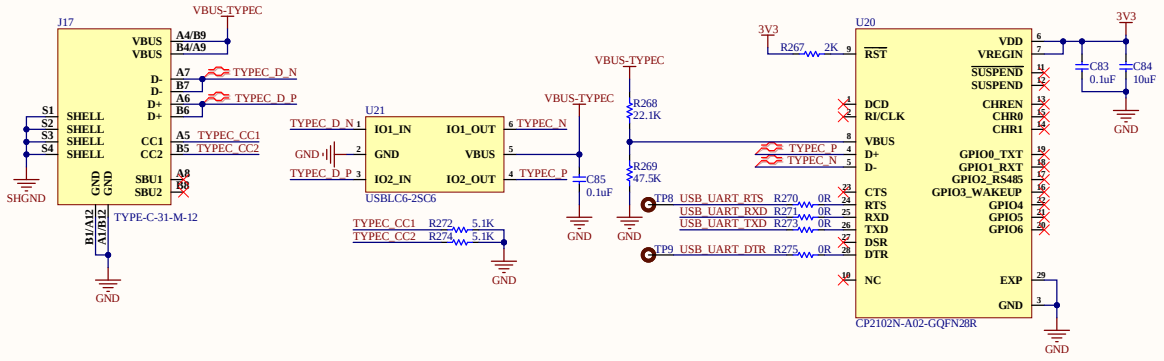
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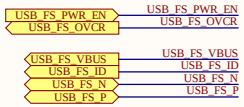
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Date: 13-Dec-24	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 15

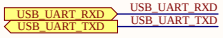
MICRO USB

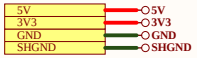


DEBUG TYPE-C





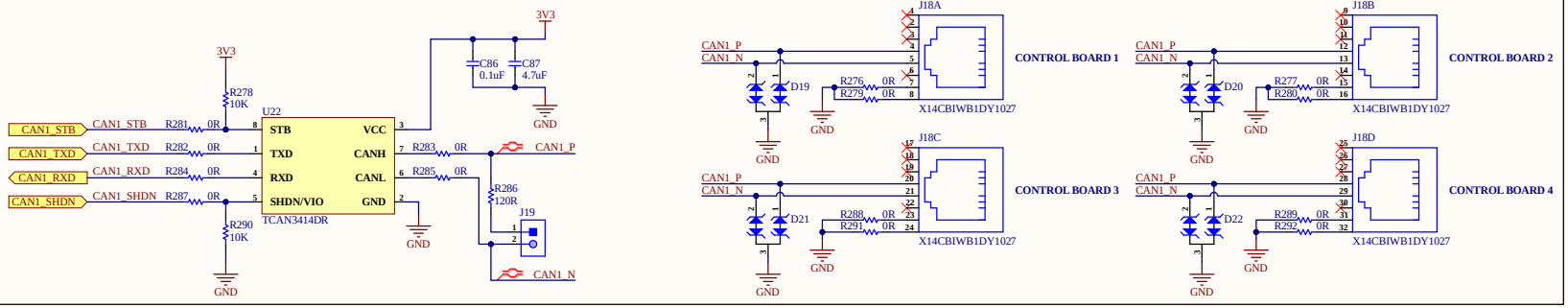




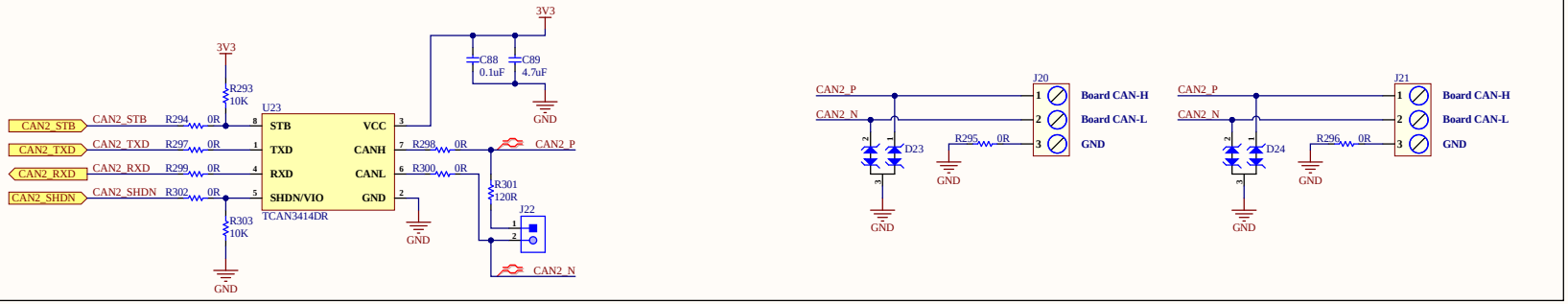
eSpark Consulting Group

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Date: 13-Dec-24	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 15

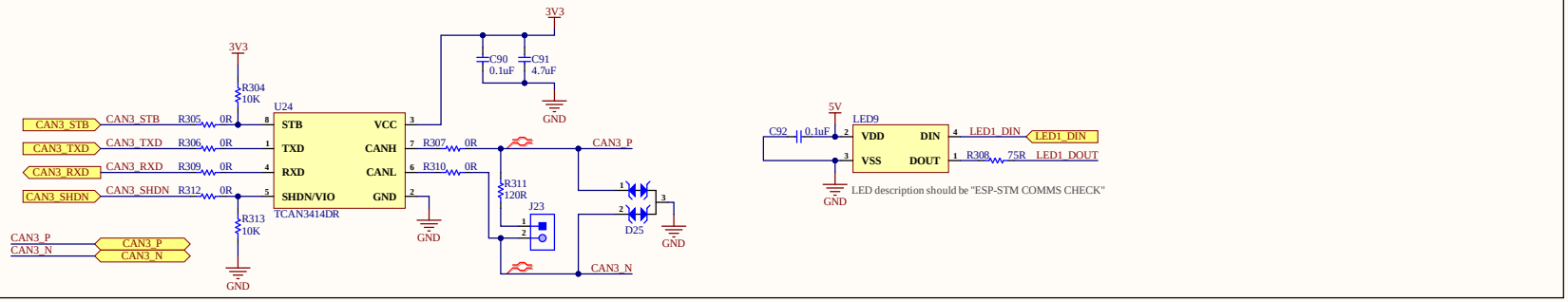
CAN 1



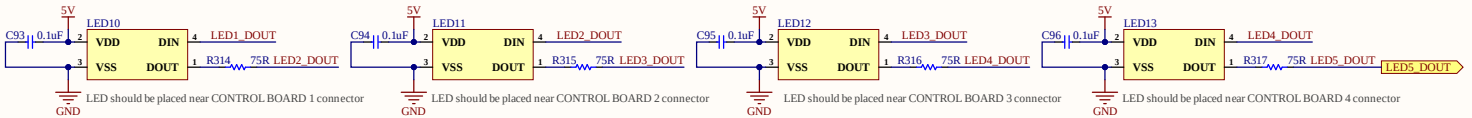
CAN 2



CAN 3



LEDs

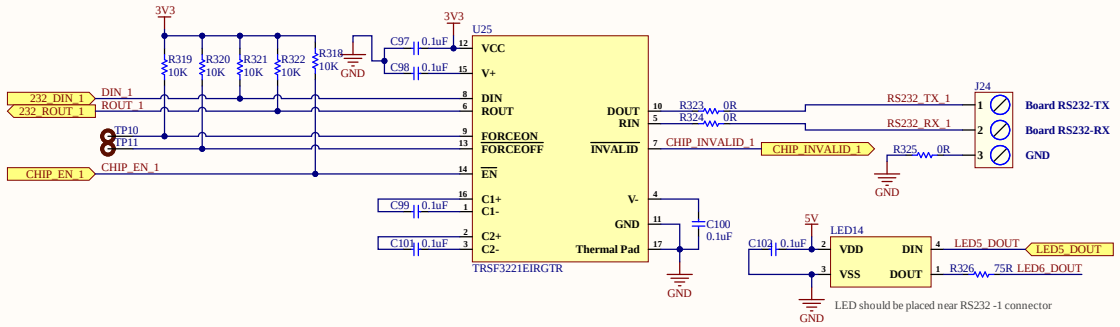


5V
3V3
GND

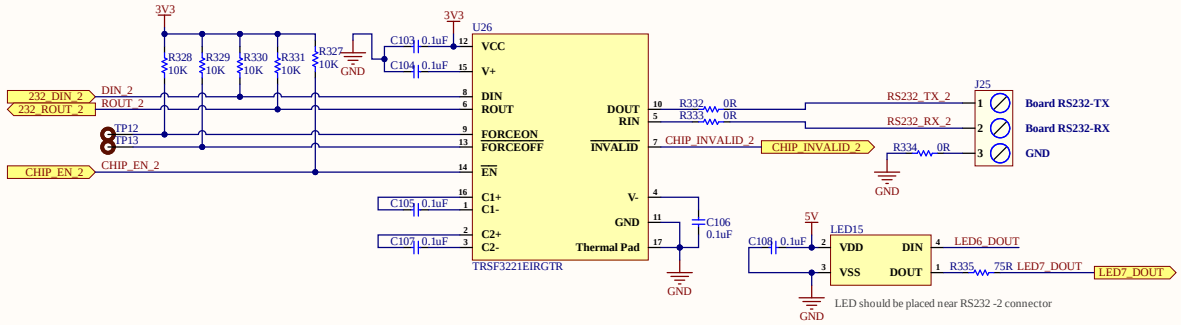
eSpark Consulting Group

Title: CAN-Sheet.SchDoc	
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Date: 13-Dec-24	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 15

RS232 - 1



RS232 - 2

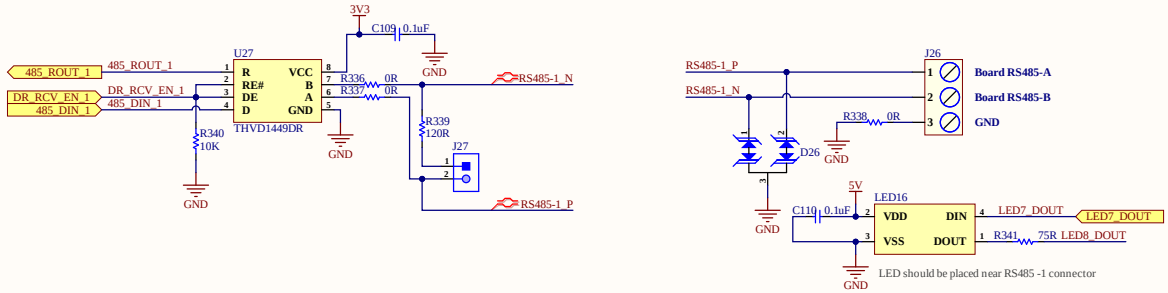




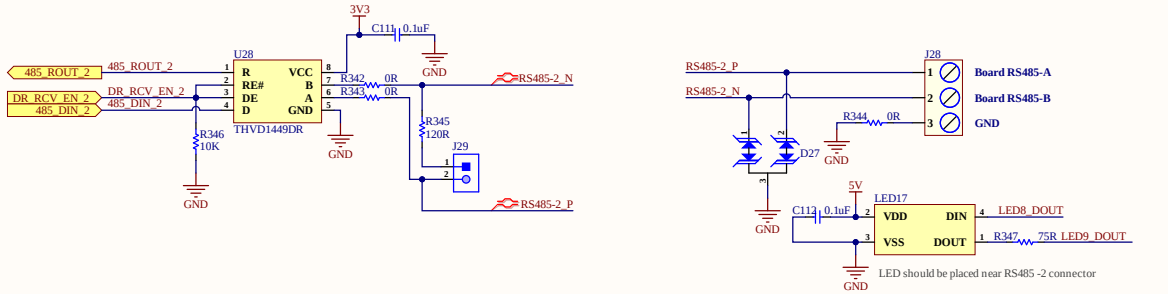
eSpark Consulting Group

Title: RS232-Sheet.SchDoc	
File: Master-Board-v2.0.PrjPcb	
Date: 13-Dec-24	Page Num: 11
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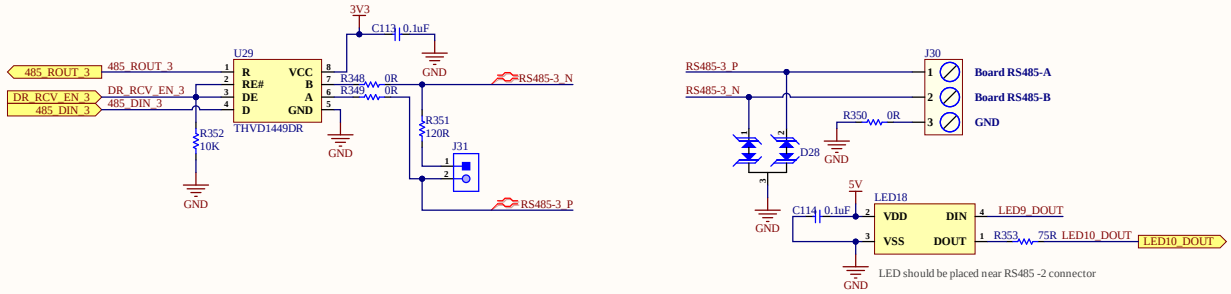
RS485 - 1

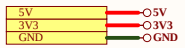


RS485 - 2



RS485 - 3

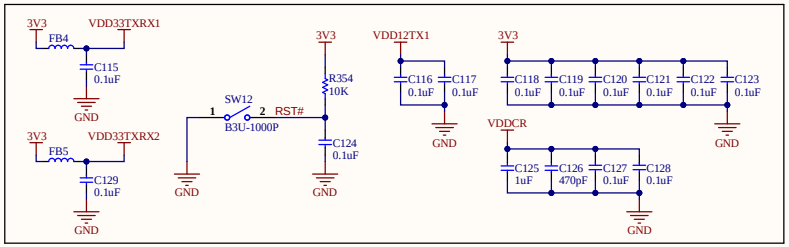


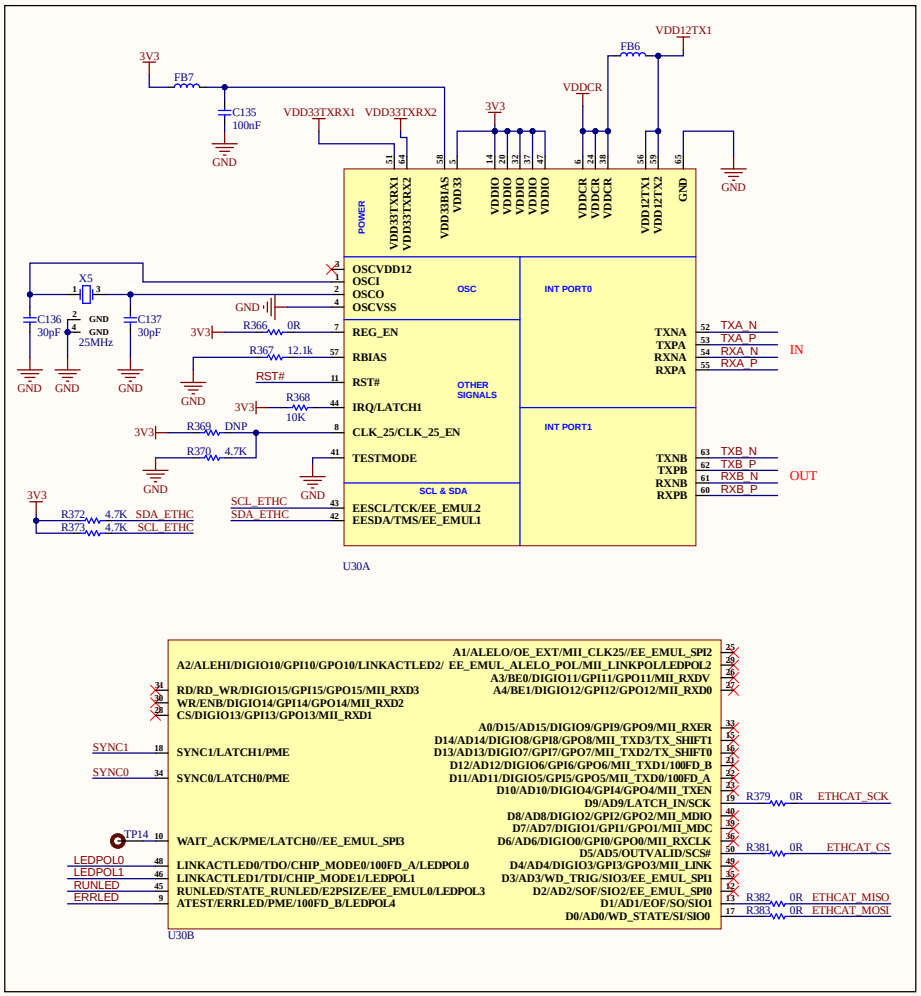


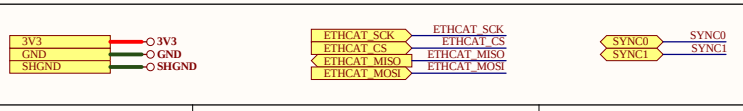
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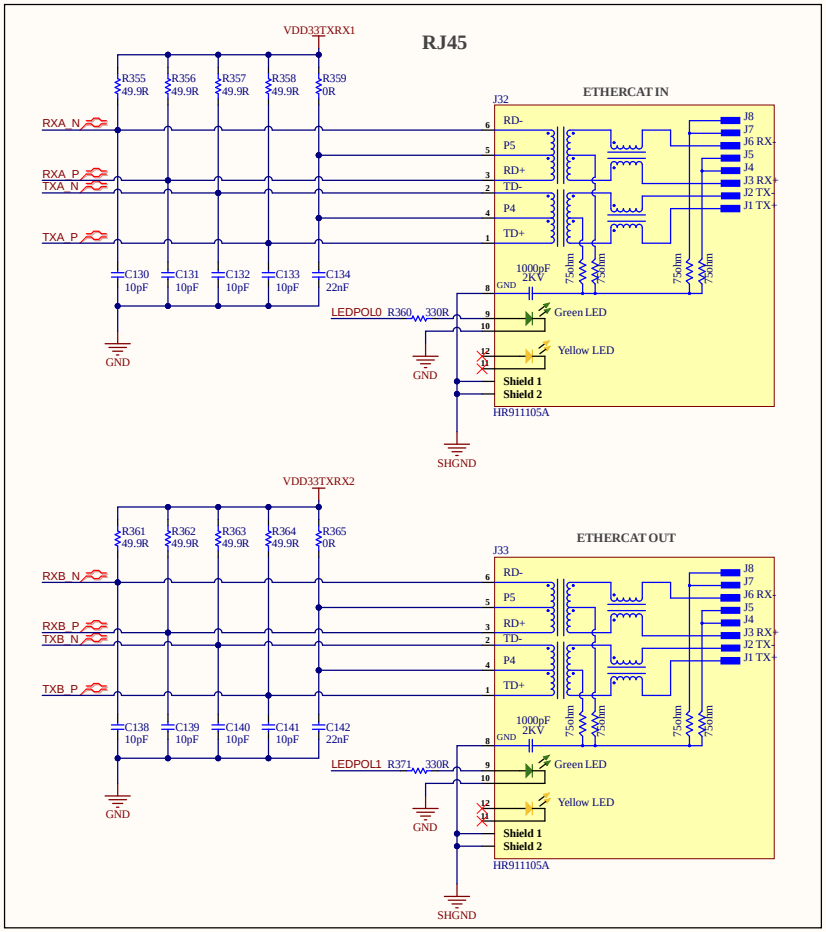
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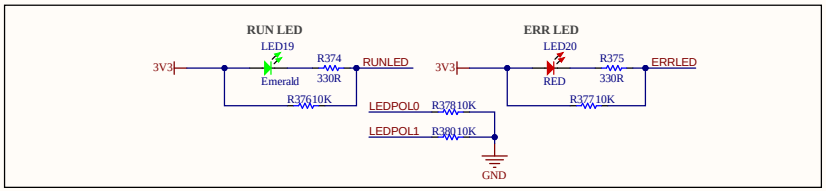
ETHERCAT

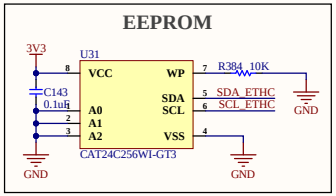








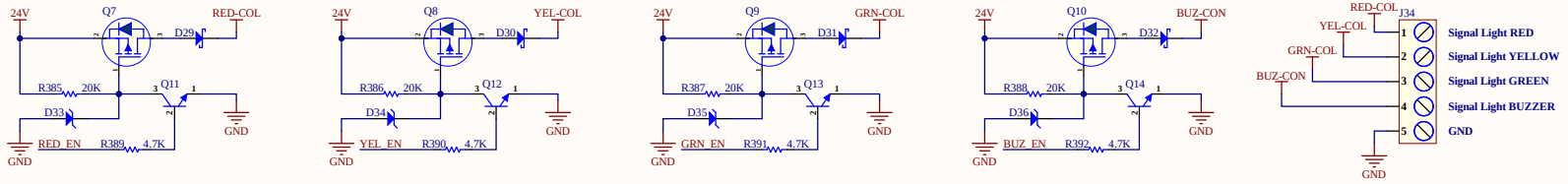




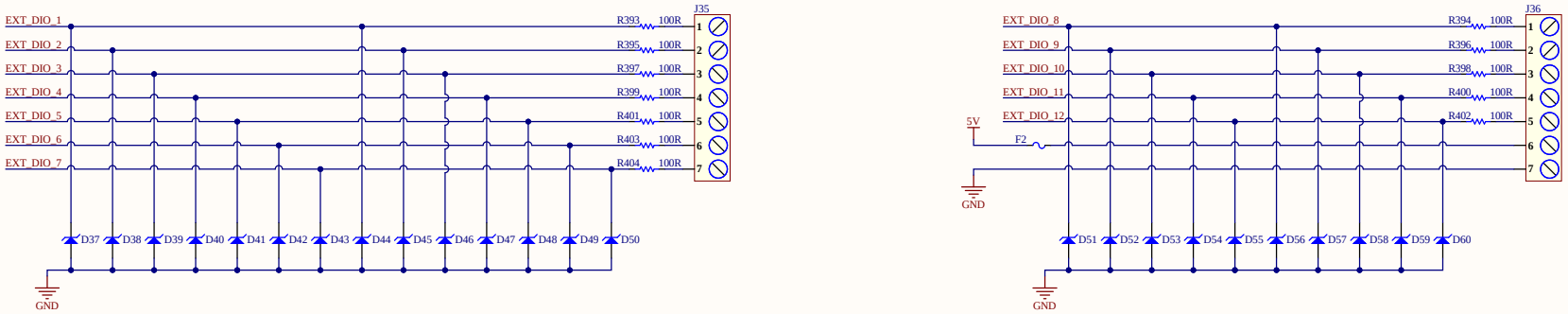
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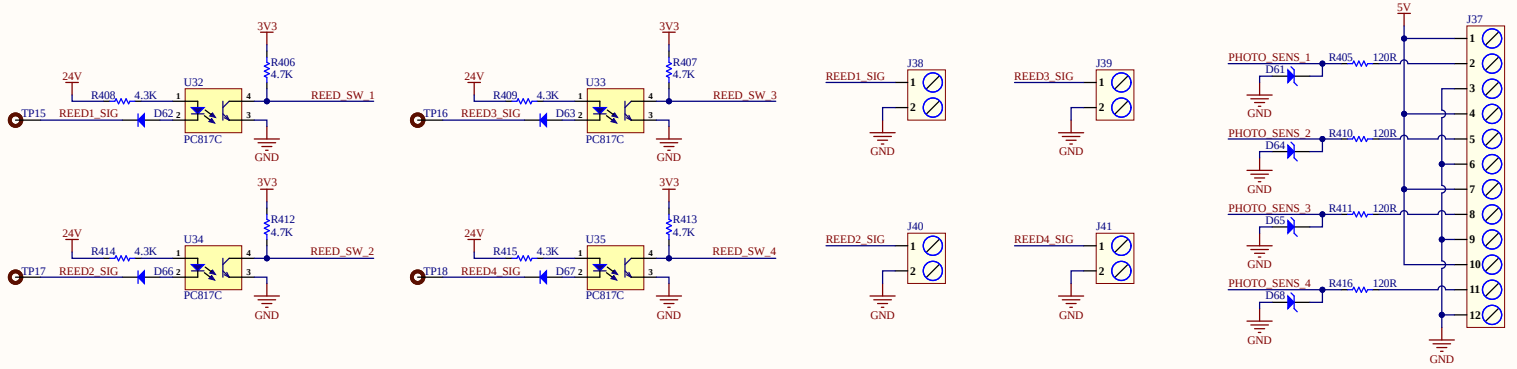
SIGNAL LIGHTS

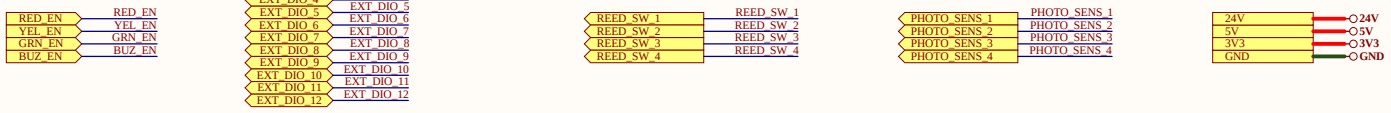


EXTRA DIOs



REED AND PHOTOELECTRIC SENSOR

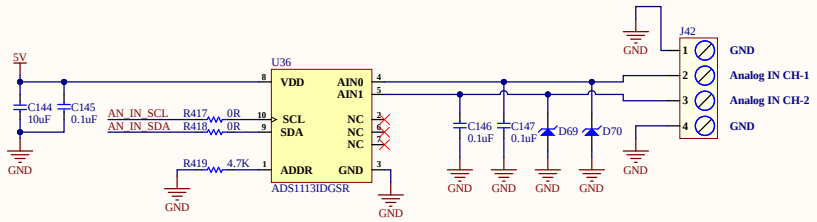




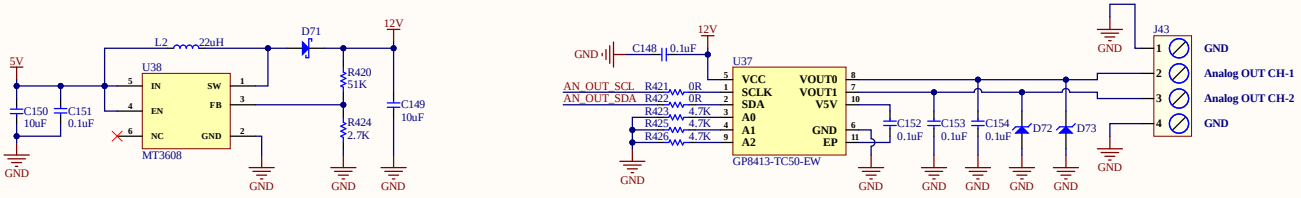
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ANALOG INPUTS 0-5V



ANALOG OUTPUTS 0-5V



AN_IN_SCL — AN_IN_SCL
AN_IN_SDA — AN_IN_SDA
AN_OUT_SCL — AN_OUT_SCL
AN_OUT_SDA — AN_OUT_SDA

5V — 5V
GND — GND

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Title: Analog-AIO-Sheet.SchDoc	
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